Case Study: Re-Engineering Thermal Limits for Next-Gen Hybrid Motors

Client: EV Manufacturers & Automotive Powertrain Engineers

Prepared by: FEAAM GmbH

Subject: Mitigating Thermal Bottlenecks in Hybrid Motors via Integrated Flux-Barrier Cooling

Executive Summary

Modern hybrid (HEV) and battery electric vehicles (BEV) require motors that are powerful, compact, and highly efficient. However, high power density creates a "thermal bottleneck" that can lead to irreversible magnet demagnetization and insulation failure. FEAAM GmbH has developed an innovative cooling approach that repurposes "flux barriers" as active thermal arteries. This design achieves a 20 K reduction in peak temperature during high-stress cycles. FEAAM helps by providing high-fidelity thermal modeling and fluid dynamics optimization, allowing manufacturers to maximize power density within the restricted packing areas of modern vehicle powertrains.

The Challenge: The Thermal Bottleneck

To meet market demand for range and power, manufacturers push motors to their absolute limits. This creates severe risks:

- Magnet Sensitivity: Permanent magnets can suffer irreversible performance loss if they overheat.
- Space Constraints: Hybrid drives offer very restricted areas for motors, making traditional cooling systems bulky and impractical.
- System Reliability: Severe thermal conditions lead to insulation failures and motor breakdown.

The FEAAM Solution: Integrated Flux-Barrier Cooling

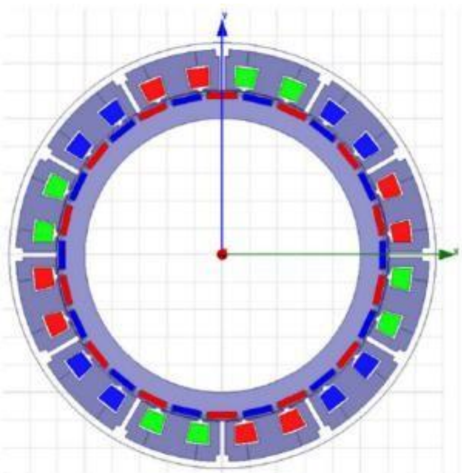

Fig. 1: Geometry of the studied PM Machine.

FEAAM transitioned from a conventional external cooling jacket to an integrated architecture using a 24-teeth/28-poles machine structure.

- Active Thermal Arteries: Instead of merely guiding magnetic flow, stator flux barriers are reimagined as high-efficiency cooling channels.
- Direct Heat Dissipation: By flowing coolant directly through these internal barriers, the cooling area is significantly increased without enlarging the motor.
- Optimized Turbulent Flow: Using Advanced Modeling and CFD, Engineers tuned geometric parameters to maximize heat extraction.

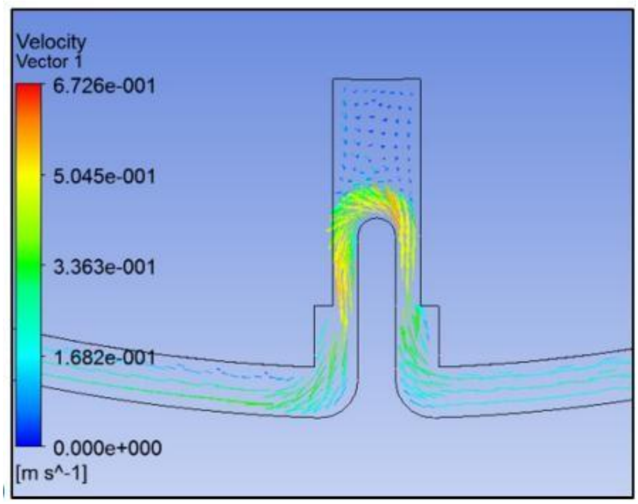

Fig. 2: Fluid flow through the final cooling channel.

Measurable Results:

FEAAM validated the new design against a standard cooling jacket using high-precision Finite Element Analysis and physical testing.

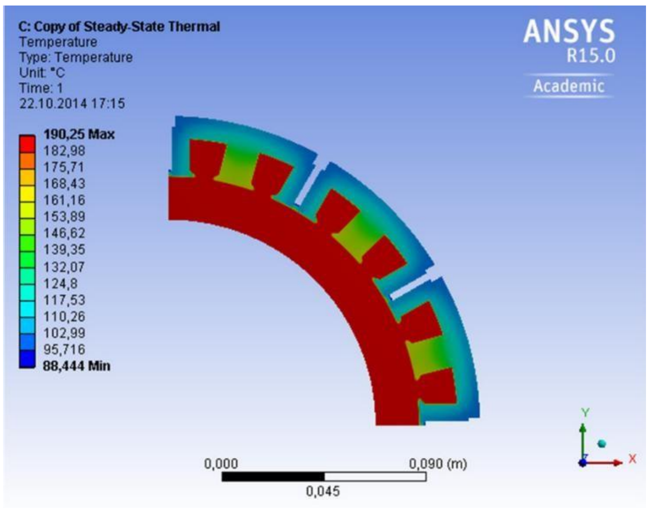

Fig. 3: Thermal Analysis for the flux barrier cooling.

Feature	Standard Cooling Jacket	FEAAM Flux-Barrier Method	Benefit
Peak Slot Temp.	210°C	190°C	20 K Safety Margin
Cooling Effect	Baseline	10% Improvement	Higher Power Density
Temp. Gradient	10 K	4 K (Optimized)	Thermal Uniformity

Fig. 4: Water with injected ink flows through the cooling jacket.

Bottom Line for Your Business

This technology provides a definitive competitive moat for manufacturers in the EV market. By choosing FEAAM's design approach, your company can:

- **Mitigate Warranty Risk:** Precision thermal mapping prevents catastrophic magnet failure and subsequent recall risks.
- **Accelerate Time-to-Market:** Virtual prototyping via CFD bypasses expensive "build-break-repeat" cycles.
- **Achieve Superior Power Density:** Deliver higher performance within harsh, compact environments that standard cooling cannot handle.

FEAAM can assist your team by reimagining stator flux barriers as active thermal arteries, utilizing advanced computational fluid dynamics (CFD) to achieve a 20 K reduction in peak temperatures and unlock superior power density within the restricted packaging areas of modern hybrid vehicle powertrains.